

3.10.1 Physics of the eye (A-level only)

3.10.1.1 Physics of vision (A-level only)

Content

The eye as an optical refracting system, including ray diagrams of image formation.

Sensitivity of the eye; spectral response as a photodetector.

Spatial resolution of the eye; explanation in terms of the behaviour of rods and cones.

Properties of converging and diverging lenses; principal focus, focal length and power,

$$power = \frac{1}{f}; \quad \frac{1}{u} + \frac{1}{v} = \frac{1}{f}; m = \frac{v}{u}$$

Myopia, hypermetropia, astigmatism.

Ray diagrams and calculations of powers (in dioptries) of correcting lenses for myopia and hypermetropia.

The format of prescriptions for astigmatism.